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Super classical quantum mechanics: The best interpretation of nonrelativistic quantum mechanics

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It has been shown that Newtonian classical mechanics (NCM) suffers from several kinds of *chaotic indeterminacies*. That means, a large set of problems treated with NCM gives results which are in wild disagreement with observation. In the present paper, these shortcomings are repaired in a simple, obvious, and essentially unique manner. The NCM theory is thereby transformed into a new theory which is fully equivalent to the Heisenberg, Schrödinger, and Dirac nonrelativistic quantum mechanics, with the vital addition of Born's probabilistic interpretation of the wave function built in from the start. I call this new theory "super classical quantum mechanics" (SCQM). Using Ehrenfest's theorem of 1927, the *classical limit* of the new theory, SCQM, is seen to give exactly the results expected of the repaired Newtonian theory of classical mechanics. © 2001 American

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I. INTRODUCTION

This approach offers enormous advantages, not only for giving the first physically reasonable interpretation of quantum mechanics, but also for its contributions to the quantum theory of measurement, and for the resolution of *all* of the so-called paradoxes of traditional nonrelativistic quantum mechanics.

A. A little bit of my early history

As a 17-year-old freshman at the University of California in Berkeley in 1930, I took a course, Physics 1A, on classical mechanics, from Victor Lenzen. It mostly dealt with Newton's second law of motion. This law seemed to work wonderfully well with objects, called "particles," sliding along smooth inclined planes or moving up and down at the end of an ideal Hooke's law spring.

In 1930, I wondered how Newton's laws of motion could give such a good description of phenomena studied in the undergraduate laboratory which was an integral part of Physics 1A. After some fruitless speculations, I decided that the most important object of physics was to study interesting laboratory phenomena, and to try to make a mathematical model in which the mathematical symbols imitated, in a way to be determined, the motions of the physical system. I regarded this as a game, to be taken seriously only if it worked well. The thought occurred to me that there might be several different mathematical models which would, more or less, correspond to the same physical behavior. If that happened, some kind of value judgment would have to be found, perhaps based on criteria such as the accuracy, beauty, and elegance of the theories.

During the next 20 years, I saw two examples of this sort of thing. In 1940, Richard Feynman, a Princeton graduate student of John Wheeler, proposed an "absorber theory" as an alternative to Maxwell's theory of electromagnetism. Another example occurred in 1951. David Bohm had received a Berkeley doctoral degree from Robert Oppenheimer in the early 1940s. For a short time before he became victim of a witch hunt, he was an assistant professor of physics at Princeton, and proposed a "hidden variable theory" of quantum mechanics. He thought this overcame some of the difficulties of quantum mechanics. In each of these cases, the new form of theory was very different from the older form, but it gave

equivalent results for simple test problems. On each occasion, I had no difficulty in making my own personal choice, which favored the older version of theory, for reasons of simplicity and beauty. However, I gave good marks to both Feynman and Bohm for trying new things, and my belief in the important role of this kind of physical model building was strengthened.

After eight years in Berkeley, with short visits to other universities at Ann Arbor, Michigan, and two closer ones: Stanford University near Palo Alto, and the California Institute of Technology in Pasadena, California, I had learned a lot from my thesis advisor, Robert Oppenheimer, and others, including William Williams, Robert Serber, Felix Bloch, Arnold Nordsieck, George Uhlenbeck, Enrico Fermi, Paul Epstein, Victor Weisskopf, J. H. Van Vleck, and I. I. Rabi. I received a Berkeley Ph.D. in Theoretical Physics in 1938. The second half of my doctoral thesis¹ gave a 1930's version of a meson theoretical description of the structure of neutrons and protons.

I liked quantum mechanics (QM) very much. The subject was hard to understand, but easy to apply to a large number of interesting problems. On balance, QM has been very good to me. I owe a lot to it. If I had not been able to understand and use the material in Hans Bethe's 1933 Handbuch der Physik article on one- and two-electron atom problems, I would never have been able to participate in the 1945–1951 experimental study of the fine structure of the hydrogen atom.

I also recognize that the experimental manipulation of simple states of atoms (one at a time), in the hydrogen fine-structure experiments, helped me formulate the vital ideas applied in my later papers on the quantum theory of measurement.

Since 1946, I have given lectures on the theory and applications of quantum mechanics at seven universities: Columbia, Stanford, Harvard, Oxford, Yale, Simon Fraser, and Arizona. Over five decades, I never understood the real meaning of QM. For various reasons, I stopped teaching the subject entirely in the mid-1980s. Since then, I have learned a lot about the foundations of quantum mechanics. I deeply wish that I could have taught the subject better in my lectures to nearly four decades of graduate students.

In the present paper, I am going to use an example of

model building for simple nonrelativistic quantum mechanical problems, like that of the fall of Newton's apple under the force of a uniform gravitational field. The moving object is taken to be a "particle" in the quaint manner used by physicists. For reasons of simplicity, I will mostly consider problems dependent on time and one spatial dimension.

II. ATTEMPTS TO GET TO THE SCHRÖDINGER EQUATION IN DIFFERENT WAYS

I have made several attempts to think of ways to get to the Schrödinger equation in a nonhistorical manner. One of these is to be found in a paper entitled "Suppose Newton had discovered the Schrödinger equation,"² which involved an imagined instance of divine intervention. Another paper, entitled "The borderland between quantum and classical mechanics,"³ relied more on human guess work. The present paper is based on an improved version of the latter method.

A. Quantum theory of measurement

A substantial amount of my work during the last 40 years has involved the quantum theory of measurement. During those years, my ideas on the subject evolved considerably. My 1969 paper⁴ in *Physics Today* came from a lecture I gave at the 1968 Conference of Physics Nobel Prize Winners, held in the German city of Lindau on Lake Constance (with Werner Heisenberg and Paul Dirac in the audience). This lecture dealt with a model for preparation of a quantum state. Measurements of position and momentum were also discussed. Measurements of some other observables made use of an idealized Stern–Gerlach meter apparatus. At that time, I did not know of the meter method (for coordinates), described in the last two pages of the last, sixth, chapter of von Neumann's 1933 book.⁵ I did know of the discussion in his third chapter of what is often called the "collapse" or "reduction" of the wave function. Very similar ideas had been expressed in the first chapter of Dirac's 1930 book on the "Principles of Quantum Mechanics." I first tried to read this book in the summer of 1932 in the Science Reading Room of the Los Angeles Public Library. I found Dirac's discussion of the collapse to be absurdly naive, and could not imagine that anybody would take it seriously. When I found that many people did take it very seriously indeed, I wrote some papers^{6–8} critical of the idea of the collapse of the wave function.

In the early 1980s, I learned from Melvin Lax (of BTL and CCNY) of the 1965 Bell System Technical Journal paper by Arthurs and Kelly,⁹ who made use of the von Neumann meter for measurement of particle position. They also devised a similar meter model for simultaneous measurement of particle momentum.

By the time of the 1986 meeting on quantum measurement theory at the New York Academy of Sciences,¹⁰ I had learned how to use, and improve upon, von Neumann's meter method for position measurements. I presented more refined methods of measurement at later meetings in 1986 at Tokyo,¹¹ at Como,¹² in 1989 at College Park, MD,¹³ and in Turin¹⁴ during 1989. In my view, these papers contained very good physics, but they never received the recognition they deserved. In my view, their most important defect was that, in discussing the measurement process, I had to change back and forth between quantum and classical mechanical descriptions of the meter measurement system. The present

paper provides a better method for dealing with this transition problem because the classical and quantum theories are now brought together into one theory in the limit in which Dirac's constant $\hbar$ becomes very small.

B. Rewriting the history of quantum mechanics

One of the purposes of the present work is to show how much simpler the history of quantum mechanics could have been, than it actually was. Teachers and students of quantum mechanics in the twenty-first century should be greatly helped by the new approach. The well-understood classical mechanics is recast into an equivalent, but more general, form. Almost automatically, the Newtonian classical mechanics (NCM) theory turns out to be an improvement on the traditional nonrelativistic quantum mechanics. It is free of most of the baggage of misunderstandings carried by the historical quantum mechanics through far too many years of the twentieth century.

C. Historical misunderstandings

One of the advantages gained by our treatment is that we can forget (if not forgive), many of the fuzzy ideas^{15–17} that cluttered the path from classical mechanics to quantum mechanics. These include: Planck's theory of blackbody radiation, Einstein's light quantum and treatment of the photoelectric effect of 1905, Bohr's 1913 quantization of atomic states, Einstein's discussion¹⁸ of radiative processes of 1917, Bohr's 1918 correspondence principle, de Broglie's waves of 1924, Heisenberg's matrix mechanics of 1925, Schrödinger's wave equation of 1925, Dirac's quantum mechanics of 1925–1926, Heisenberg's uncertainty principle of 1927, Bohr's principles of complementarity and wave-particle duality, Dirac's and von Neumann's notions of the collapse of wave packets, and the excessive use of Hilbert space in mathematically inclined papers. We would have been spared much of the agony needlessly suffered from the two-slit diffraction-pattern paradoxes, the Einstein–Podolsky–Rosen paradoxes, and the Schrödinger cat paradox.^{19,20} Also, unnecessary were Bohm's quantum mechanics, hidden variable theories, and Bell's inequalities. Discussions of these and other variations abound in the pages of *Physics Today*, *Physics World*, *New Scientist*, and *Scientific American*. Many more are to be found on web screens of Netscape and Internet Explorer.

In Ref. 16, one finds many abbreviations like DH (decoherent histories), with words like "reality" and "nonlocality," SL (spontaneous localization), and BQM (Bohmian quantum mechanics). It is widely rumored that the editors of the *Physical Review* and other journals find that papers on the principles of quantum mechanics, or its measurement theory, are too troublesome to get properly refereed. I feel strongly that papers on teleportation are beneath contempt. Delayed choice experiments²¹ have never been adequately discussed using any systematic theory of quantum mechanical measurement. The same can be said of quantum computers and quantum cryptography.²² The bulk of papers on these subjects lack any understanding of the intrinsic probabilistic nature of quantum mechanics. On the other hand, my four papers^{10–12,14} make use of computerized random number generators to simulate the quantum fluctuations introduced by the use of meters in the measurement process. Anyone who wants to use a quantum computer ought to know that he

has built in to it something like a coin tosser as a circuit element. That would not be a desirable feature for a von Neumann machine.

In my opinion, the above troubles can easily be avoided if we derive the probabilistic features of quantum mechanics properly by “repairing” the nondeterministic Newtonian classical mechanics. The resulting theory contains Dirac’s constant $\hbar \cong 10^{-34}$ J second with dimensions of an “action,” i.e., an energy multiplied by a time (or a distance multiplied by a momentum). Physical problems which are characterized by actions very large compared to $\hbar$ can, in suitable cases, be treated by classical mechanics, and problems with smaller action will automatically be dominated by quantum features.

III. WHAT ARE THE TROUBLES OF CLASSICAL MECHANICS?

We have been claiming that NCM is *undeterministic*. Starting with the fourth paragraph below, we describe several types of nondeterministic behavior. Consider the motion of a particle of mass m moving along a frictionless straight line under the action of a force F . The particle’s position x is a function $x(t)$ of time t . Its velocity v is $v(t) = dx(t)/dt$, its acceleration a is $a(t) = dv(t)/dt = d^2x(t)/dt^2$. In a typical application, the force F might be a function of time, $F(t)$, or a functional $F(x(t), t)$ of the function $x(t)$ and the time t . For the case of a potential energy field $V(x, t)$, one would have a force F given by

$$F(x, t) = - \frac{\partial V(x, t)}{\partial x}.$$

Extensions to many-body problems are fairly straightforward. Newton’s classical equation of motion is

$$m \frac{d^2x}{dt^2} = - \frac{dV}{dx}.$$

There are some simple analytically solvable one-dimensional problems for potential functions such as $V(x) = \text{constant}$, x , x^2 , and with more trouble, x^4 , etc. Thus, for $V(x)$ proportional to x we get uniformly accelerated motion; for a quadratic, x^2 , we get trigonometric motion. For a quartic potential, x^4 , we are led to elliptic functions.

Examples of the simplest potential energy functions are:

- (1) $V(x) = 0$, for a free particle at x , moving along the x axis;
- (2) $V(x) = mgx$, for a body falling in a uniform gravitational field; and
- (3) $V(x) = (1/2)kx^2$, simple harmonic oscillator, with Hooke’s law restoring force exerted by an ideal spring.

In these cases, Newton’s equation can be solved in terms of elementary functions. For most potential functions, no analytic solutions can be found, and one must have recourse to numerical integration, preferably with help from a computer.

IV. DISCUSSION OF THE DETERMINISM OF CLASSICAL NEWTONIAN MECHANICS

Some important questions arise. How deterministic is the theory based on Newton’s equations? What is its range of

validity? For how long a time can one expect to have a valid description of the motion of a system? In any case, the longer the time elapsed, the more the motion will be disturbed by remote objects or by forces originating from outside the system.

It might seem possible to give a pretty good treatment of long time motion if the system of interest is well isolated from the rest of the universe. Then one might expect to have a high degree of determinism. However, in his Nobel Prize lecture of 1954, Max Born²³ pointed out a mismatch between mathematics and physics. (I had no thoughts of this idea in 1930.) In mathematics, numbers are imagined to be capable of infinitely high precision. On the other hand, in physics, the position of a particle can only be specified by numbers of finite precision. This mismatch might not seem too troublesome for problems of interest, except after very long times.

There is another point that Born might have made, but did not: The Newtonian concept of a particle was only a convenient fiction. There probably are no elementary particles anywhere in nature. Galileo had a lamp, swinging in a cathedral, or a block of wood sliding down an inclined plane, or a stone or a piece of paper dropped from the Tower of Pisa. Newton might have had an apple or a heavenly body. He probably realized that these objects were composite structures, but did not follow that idea further. Electrification of some bodies by friction was known, if not much studied in the 17th century. The discovery of electrons required several centuries more.

In Newtonian mechanics, there are two basic quantities used to characterize the motion of a particle: a coordinate $x(t)$ and a momentum

$$p(t) = mv(t) = m \frac{dx(t)}{dt}.$$

These involve very different physical concepts. However, it was tempting to treat them as equally important elements of the theory. This tendency was much encouraged by the beautiful transcription of Newton’s equations of motion into their Hamiltonian form in the nineteenth century. However, I think that finding a better formulation of quantum mechanics is even a more important task.

There are certain problems involving classical measurement and state preparation: How do we know that a particle is at $x=0$ at $t=0$? Or how do we know that a particle has a velocity v at time $t=0$? In the 1660s, the position of a particle could be measured with a microscope having a suitable distance scale. It took three centuries, from about 1660 to a bit beyond 1960, before velocity could be measured with a police Doppler laser speed trap apparatus. Both of these instruments are optical instruments, and cannot be readily modeled with the laws of mechanics. At the time of Newton, position could be measured, but velocity or momentum determination would ordinarily have required two position measurements at closely separated times.

It would be nice to have a mechanical model for the measurement processes of classical mechanics which could be analyzed using only the laws of Newtonian mechanics. Heidi Fearn and I²⁴ have given an indication of how position measurements can be carried out in Newtonian mechanics. My papers on the quantum theory of measurement^{10–12,14} are, in effect, just quantum transcriptions of such classical discussions. Later papers will return to this problem.

V. WHAT ARE SOME TROUBLES WITH THE DETERMINISM OF NEWTONIAN CLASSICAL MECHANICS?

Consider the free motion of a particle on a straight line. Suppose that we want to have its initial position to be $x(0) = 0$ and its initial velocity $v(0) = [dx(t)/dt]_0 = 0$. The traditional Newtonian result is that x would remain at $x(0) = 0$ for all later times. However, in a physical world, both of the starting values of $x(0)$ and $[dx(t)/dt]_0$ would have values somewhat out of our control. For a long term solution, we might ignore the effect of a slight offset in position $x(0)$, but a very small value of the initial velocity $v(0)$, when multiplied by a large value of elapsed time, could lead to a flagrant and highly unphysical discrepancy between theory and experiment for evaluation of $x(t)$. Hence, the Newtonian treatment needs to be “repaired.”

Another example: Consider a particle, initially near a position $x=0$, acted upon by a force derived from a peaked potential energy function $V(x)$, involving a curve called the Witch of Agnesi:²⁵

$$V(x) = \frac{V_0}{1 + \frac{x^2}{x_0^2}},$$

where the peak potential energy magnitude V_0 is positive, and the distance parameter x_0 gives a measure of the peak width. For coordinate x values much smaller in magnitude than x_0 , this potential function $V(x)$ is similar to an “upside down” parabolic function of x . If the initial x value were exactly zero, a long term solution of Newton’s equation would be $x(t) = 0$. To try to confirm this experimentally, we might use the best efforts of a highly skilled person to put the particle at rest with $x=0$ at $t=0$. Even so, the long term behavior of the system will be very strongly dependent on the initial values of x and v . When I was younger, this state of affairs was called unstable equilibrium. Now, it is one of the simplest examples of a chaotic problem.

There are many other problems in Newtonian mechanics that have similar bifurcation features. One of the first to meet them, without much contemporary notice being taken, was Maxwell, in his kinetic theory of gases (~1865). A good deal of the trouble that Boltzmann had in a conflict between mechanics and thermodynamics came from similar chaos. Just think of the chaotic consequences associated with nearly head-on collisions in a gas of two ideally spherical molecules. See also Maxwell’s 1871 Inaugural Lecture²⁶ as Cavendish Professor of Experimental Physics at Cambridge. Such topics also occur in the 1896 work on celestial mechanics by Poincaré,²⁷ and in my lecture, “Quantum Chaos and the Theory of Measurement”²⁸ at the 1983 Como Conference on Quantum Chaos, and in a 1986 Royal Society Lecture by the late Sir James Lighthill,²⁹ with a title “The Recently Recognized Failure of Predictability in Newtonian Dynamics.” (In 1970, Lighthill succeeded Dirac as Lucasian Professor of Physics at Cambridge before moving to London after about 10 years. He recently died in a swimming accident.)

Other examples of such problems can be found in a 1918 book by Georg Duffing³⁰ in his treatise on nonlinear mechanics. As one goes to problems of higher dimensionality,

most classical dynamical systems have chaotic behavior, as in the long term motion of particles in the Fermi–Pasta–Ulam³¹ and Toda³² lattices.

VI. LET’S TRY GOING INTO THE x, p PHASE SPACE

I draw from the previous paragraphs the conclusion that classical mechanics should not be formulated in terms of the ordinary differential equations of NCM which determine trajectories as $x = x(t)$. Instead, one should have a probabilistic description of some kind. I have in mind the early 1900’s work of Gibbs on statistical mechanics. Consider also the 1949 Waynflete lectures of Born³³ at Magdalen College, Oxford. He moved from a trajectory approach with $x(t)$ in coordinate space to a description in a two-dimensional phase space with coordinate x , and momentum p . He made use of the probabilistic concept of a phase space density $\rho(x, p, t)$. This density $\rho(x, p, t)$ obeys the Liouville equation

$$\frac{\partial \rho(x, p, t)}{\partial t} = \{H(x, p, t), \rho(x, p, t)\},$$

involving the Poisson bracket expressions

$$\{H, \rho\} = \frac{\partial H}{\partial x} \frac{\partial \rho}{\partial p} - \frac{\partial H}{\partial p} \frac{\partial \rho}{\partial x},$$

formed from $\rho(x, p, t)$ and the Hamiltonian function

$$H(x, p, t) = \frac{p^2}{2m} + V(x, t).$$

This kind of approach would put the usual Newtonian results in probabilistic form. It could also have helped Born overcome his arithmetic precision problems. It makes use of an x, p phase space, but does not recognize that Newtonian coordinates and momenta really have enormous physical differences. Moreover, such a road does not lead to quantum mechanics.

VII. PROBABILITY DENSITY IN COORDINATE SPACE

We return to the Newtonian treatment, with the idea of using a probability density only for the time-dependent particle position coordinate x , as in a “position pointing function” $W(x, t)$. As a first guess, one could try to find a mathematical equation determining $W(x, t)$. This might have to be some kind of partial differential equation since W has to be a function depending on two independent variables x and t . For example, the equation

$$\frac{\partial W(x, t)}{\partial t} + v \frac{\partial W(x, t)}{\partial x} = 0,$$

could describe the motion of a probability density function $W(x, t)$ in free-particle motion with constant velocity v . The general solution of this equation is $W(x, t) = f(x - vt)$ where f is a normalized, but otherwise arbitrary, function of its argument, $x - vt$. The spatial dependence of the function $W(x, t)$ would determine the probability distribution for the x location of the particle at time t . A well-localized free particle at time t would be described by a function $W(x - vt)$ which was strongly peaked in its argument $x - vt$.

Of course, our object is to be able to describe more complicated motions for which forces are acting on the particle. At this point, we have a possible clue: We should look for a partial differential equation for $W(x,t)$ for cases where forces are acting on the particle.

However, solutions of such a partial differential equation could easily become negative for some values of x and t . Unlike some more famous people, I do not wish to introduce a nonsensical concept like negative probability into physics. A simple way out of the difficulty is to use a (possibly complex) probability amplitude function $\psi(x,t)$ and to assure that $W(x,t)$ remains positive by setting $W(x,t) = |\psi(x,t)|^2$. Then we can try to find a partial differential equation for $\psi(x,t)$.

VIII. GETTING A PARTIAL DIFFERENTIAL EQUATION FOR $\psi(x,t)$

Looking for a partial differential equation to determine $\psi(x,t)$ might seem like looking for a needle in a haystack. However, it turns out that two guiding principles can take us a long way to a unique solution. These are (1) simplicity, and (2) dimensional correctness. In an earlier paper,³ I have shown that this approach leads almost uniquely to the Schrödinger equation and Born's probability interpretation of the ψ function. To save space in this paper, I skip this derivation here and assume that the Schrödinger equation (complete with its "i" and " $\hbar$ "),

$$-\frac{\hbar}{i} \frac{\partial \psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x,t)}{\partial x^2} + V(x,t) \psi(x,t),$$

is already known from Ref. 3, together with the probabilistic interpretation of $\psi(x,t)$. Of course, Schrödinger did guess to get his equation, but he never understood what it meant. I strongly recommend that the function $\psi(x,t)$ should be called a " $\psi(x,t)$ function" or "a probability amplitude function" with NO mention of a (Schrödinger or Dirac) wave function or wave equation. This advice might help to reduce what I like to call unnecessary "dueling" between waves and particles. Note that $\psi(x,t)$ has no direct physical significance. Its only purpose is to make possible the calculation of $W(x,t)$. I am reminded that, in mathematics, the complex number "i" has no direct physical significance, but it may nevertheless be used to make calculations of quantities that do have such significance.

The ψ function can be normalized by providing it with a multiplicative normalization factor so that the integral

$$\int dx W(x,t) = \int dx |\psi(x,t)|^2,$$

is unity when the integration over x runs from $x = -\infty$ to $x = +\infty$.

Just as in the historical development, we can derive a probability conservation law for ψ from the ψ equation. We can also calculate the time rate of change of the time-dependent x -centroid $\langle x(t) \rangle$ or other functions of x . The centroid of the x distribution is

$$\langle x(t) \rangle = \int dx x W(x,t) = \int dx \psi^*(x,t) x \psi(x,t),$$

and the corresponding time rate of change of $\langle x(t) \rangle$, i.e., velocity, is

$$\langle v(t) \rangle = \frac{d}{dt} \langle x(t) \rangle = \frac{d}{dt} \int dx \psi^*(x,t) x \psi(x,t).$$

The time differentiation is now taken under the x integration sign. Use is made of the ψ and ψ^* equations, and some partial integrations are performed assuming that the functions ψ and ψ^* vanish as x approaches infinity. The result is a time-dependent velocity

$$\langle v(t) \rangle = \frac{d}{dt} \langle x(t) \rangle = \left(\frac{1}{m} \right) \int dx \psi^*(x,t) \frac{\hbar}{i} \frac{\partial \psi(x,t)}{\partial x}.$$

From this we see that an operator p for the momentum in the x direction, familiar in conventional quantum mechanics, can be taken as $p = (\hbar/i)(\partial/\partial x)$, which corresponds to the equation $p = m(dx/dt)$ of classical mechanics. At this stage, we could, if desired, begin to work with noncommuting x , p , and other operators, as in the familiar forms of quantum mechanics.

A further differentiation with respect to time t of the above integral expression for $\langle v(t) \rangle$ and use of the ψ equations for the time derivatives of ψ and ψ^* , and some more partial integration, leads to the 1927 Ehrenfest³⁴ theorem

$$m \frac{d^2 \langle x(t) \rangle}{dt^2} = - \int dx \psi(x,t)^* \frac{\partial V(x,t)}{\partial x} \psi(x,t),$$

which is the quantum version of the second law of motion of NCM when the "spatial width" of $\psi(x,t)$ is sufficiently small. This equation is vital for an understanding of how classical mechanics emerges from quantum mechanics in the classical limit.

It may be well to point out that Ehrenfest's theorem is here derived as an integral relationship from the probability amplitude partial differential equation and, hence, it does not necessarily have any connection to the classical mechanics of a particle. To get such a connection, we must have a situation where $\psi(x,t)$, $\psi(x,t)^*$, and hence, $\psi(x,t)^* \psi(x,t)$, are sufficiently "narrow" functions of x . Even then, in many cases the width of the ψ function may increase in time, so that the application to classical mechanics is no longer valid. Further discussion of such matters will appear elsewhere.

IX. HOW WE CAN DEAL WITH NEWTONIAN MECHANICS: LET'S TEST THIS OUT WITH OUR THREE SIMPLE PROBLEMS

It remains to be seen how well the ψ equation can imitate the results usually obtained from Newton's equation of motion for the three problems listed above.

The use of a time-dependent Wentzel, Kramers, and Brillouin (WKB) method will prove helpful. We substitute for the probability amplitude function $\psi(x,t)$ an exponentiated expression

$$\psi(x,t) = \exp \left(-\frac{S(x,t)}{\hbar} \right),$$

into the probability amplitude equation and get a time-dependent Riccati equation for $S(x,t)$,

$$-i \frac{\partial S}{\partial t} = -\frac{1}{2m} \left(\frac{\partial S}{\partial x} \right)^2 + \left(\frac{\hbar}{2m} \right) \left(\frac{\partial^2 S}{\partial x^2} \right) + V,$$

for the complex “action,” or “Hamilton–Jacobi function,” $S(x,t)$.

X. IF IT WEREN'T FOR STATIONARY STATES, WE WOULD NOT HAVE QUANTUM MECHANICS

Most of the applications of the WKB approximation in well-known quantum mechanics text books, for instance,³⁵ aim at finding approximate solutions for stationary state ψ functions. The left-hand side of the time-dependent Riccati equation is replaced by the product of the energy eigenvalue of the stationary state E and the action function $S(x,t)$. Then, with suitable approximations, one gets the usual WKB expressions for classically allowed and forbidden parts of the eigenfunction of a stationary state with energy E . With the Kramers–Airy function, or the Kramers–Zwaan connection, one gets good approximations for energy eigenvalues. The emphasis on stationary states in quantum mechanics came historically from the spectroscopic studies of the nineteenth century.

Most traditional text books on quantum mechanics³⁵ deal with steady, or stationary, states. Time-dependent, moving ψ packet solutions seem too hard to find. This widens the gap between classical and quantum mechanics, and has seriously undermined the unity of the sciences of mechanics.

XI. WE WANT TO HAVE TIME DEPENDENT SOLUTIONS OF QM PROBLEMS

In the present work, we want to get solutions for the time-dependent probability amplitude packet for problems (1), (2), and (3) which have different powers 0, 1, and 2 of x in their potential functions. The Riccati equation contains a second-order differentiation of S with respect to x , and the square of a first-order differentiation of S with respect to x . As a result, we can get simple exact particular solutions of a wide class of time-dependent ψ equations. To bring the various possibilities into a compact form, we consider a potential function $V(x,t)$ in the form of a sum of a time-dependent linear and a time-dependent quadratic (parabolic) function of x ,

$$V(x,t) = V_1(t)x + V_2(t)x^2$$

or more explicitly,

$$V(x,t) = -F(t)x + \frac{1}{2}m\omega(t)^2x^2.$$

For such a parabolic polynomial in x , the action function $S(x,t)$ will also be a quadratic polynomial in x . For potentials V with other kinds of x dependence, we will soon be indicating an efficient way to get useful approximate time-dependent solutions of ψ packet problems.

The quantities $F(t)$ and $\omega(t)$ have simple intuitive interpretations, and can have arbitrary time dependence. A way of describing the notation is to say that the force $F(t)$ characterizes the local slope of the potential energy $V(x,t)$, i.e., represents a spatially uniform, but time variable, force field $F(t)$, and the coefficient of the x^2 term corresponds to the local x^2 curvature of the potential function $V(x,t)$, i.e., we may speak of a time-dependent Hooke's law spring force constant.

XII. METHOD OF A COMPLEX GAUSSIAN ψ FUNCTION

A quite general expression for the solution of our Riccati equation for the action function S is then of the form

$$S(x,t) = \alpha(t)x^2 - 2\beta(t)x.$$

This choice of the action function makes ψ into a complex Gaussian function. The quantities $\alpha(t)$ and $\beta(t)$ are complex functions of the time and are still to be determined. The action function $S(x,t)$ could also have a term in x^0 , which would serve to renormalize the probability amplitude function ψ . I prefer to work now with un-normalized ψ functions, and will be careful to allow for proper normalization when calculating expectation values. Hence, the x^0 terms will now be omitted from $S(x,t)$.

For this $S(x,t)$ to be a solution of the Riccati equation, it is necessary that $\alpha(t)$ and $\beta(t)$ obey the two coupled non-linear ordinary differential equations

$$\begin{aligned}\frac{d\alpha(t)}{dt} &= i\left(V_2(t) - \frac{2\alpha(t)^2}{m}\right), \\ \frac{d\beta(t)}{dt} &= i\left(V_1(t) - \frac{2\alpha(t)\beta(t)}{m}\right).\end{aligned}$$

There is far more generality in these two equations than we will need, since they are still valid when F , ω , and m , and even $\hbar$, can be arbitrary functions of time. In earlier years, one could have solved the differential equations in simple but important cases. Nowadays, one might use computer analysis even for the soluble cases.

To sum up, for the class of potentials with linear [and (or)] quadratic x dependence, there are relatively simple complex Gaussian ψ function solutions. As discussed below, these can be used as a starting point for approximate solutions of a much wider class of x -dependent potentials.

A. Free particle

Case (1) is that of a free particle. F and ω are both zero. The equation for $d\alpha(t)/dt$ does not depend on $\beta(t)$: We have

$$\frac{d\alpha(t)}{dt} = -2i\frac{\alpha(t)^2}{m},$$

with solution given by

$$\frac{1}{\alpha(t)} = \frac{1}{\alpha_0} + \frac{2i\hbar t}{m},$$

or

$$\alpha(t) = \frac{\alpha_0}{1 + \frac{2i\hbar\alpha_0 t}{m}},$$

where α_0 is the initial value of $\alpha(t)$ at $t=0$. The corresponding equation for $d\beta(t)/dt$ is

$$\frac{d\beta(t)}{dt} = -i\left(\frac{2\alpha(t)\beta(t)}{m}\right).$$

One solution is $\beta(t)=0$. The general solution for $\beta(t)$ is a constant multiple of $\alpha(t)$. If the constant is real, the particle

centroid remains at rest. If the constant is pure imaginary, the centroid moves with constant velocity.

A free particle $\psi(x,t)$ packet solution is

$$\psi(x,t) = \exp(-\alpha(t)x^2 + 2\beta(t)x).$$

The corresponding probability density is

$$W(x,t) = |\psi(x,t)|^2 = \exp(-2 \operatorname{Re}(\alpha(t))x^2 + 4 \operatorname{Re}(\beta(t))x).$$

The centroid of this Gaussian exponential distribution is at

$$\langle x(t) \rangle = \frac{\operatorname{Re}(\beta)}{\operatorname{Re}(\alpha)}.$$

That this packet moves with uniform velocity is also easily seen from the integrals in the Ehrenfest relation for a free particle.

B. Uniform acceleration in a constant uniform force field

With a constant $F = mg$ (and $\omega = 0$), the $\alpha(t)$ is just that for a free particle, and $\beta(t)$ obeys the easily solved equation

$$\frac{d\beta(t)}{dt} = -i \left(mg + \frac{2\alpha(t)\beta(t)}{m} \right).$$

Again, an application of Ehrenfest's theorem shows that in the classical limit of $\hbar$ approaching zero the centroid has the correct constant acceleration g .

C. Simple harmonic oscillator

Case (3) is that of a simple harmonic oscillator. Again, the use of Ehrenfest's theorem shows that, in the classical limit, the centroid $\langle x(t) \rangle$ has the correct Hooke's force law motion.

There is an important special case when $\alpha(t)$ is a constant. According to the differential equation for $d\alpha/dt$ this occurs when $\alpha = (1/2)(m\omega/\hbar)$. This α value is that describing the mean-square width of the Gaussian ground state of the simple harmonic oscillator potential function for the values of m , ω , and $\hbar$. With use of the complex Gaussian method, one could, much earlier and more easily, have found the time-dependent states discovered by Schrödinger³⁶ in Zurich in 1926, generalized by Earle Kennard³⁷ of Cornell in 1927, and called "coherent" by Roy Glauber³⁸ of Harvard in the 1960s. The time-dependent wave packet solutions for a harmonic oscillator have been given in a book on laser physics.³⁹

XIII. SOME PROPERTIES OF A COMPLEX GAUSSIAN ψ FUNCTION

The (un-normalized) ψ function is $\psi(x,t) = \exp[-S(x,t)/\hbar]$. The un-normalized probability density $WU(x,t)$ is

$$WU(x,t) = \exp\left(-\frac{S(x,t) + S(x,t)^*}{\hbar}\right).$$

With the action function $S(x,t) = \alpha(t)x^2 - 2\beta(t)x$, we get $WU(x,t) = \exp(-2 \operatorname{Re}(\alpha)x^2 + 4 \operatorname{Re}(\beta)x)$. The peak of the probability distribution is at $x_p = \operatorname{Re}(\beta)/\operatorname{Re}(\alpha)$. The standard deviation is $\sigma = 1/\sqrt{2 \operatorname{Re}(\alpha)}$.

A. Nonparabolic potentials

It would seem at first sight that one could only treat problems with potential functions made up of linear and quadratic powers of x . However, in many cases, a very good approximation can be used for a much wider class of potential functions $V(x)$. At each temporal stage of the integration procedure, the values of $\alpha(t)$ and $\beta(t)$ determine the peak coordinate x_p of the Gaussian probability distribution function $W(x,t)$, which occurs at

$$x_p = \frac{\operatorname{Re}(\beta)}{\operatorname{Re}(\alpha)}.$$

The corresponding inverse squared width parameter is $\sigma = 2 \operatorname{Re}(\alpha)$. One could then make a Taylor's expansion of the potential function $V(x)$ around the point $x = x_p$. We keep only linear and quadratic terms in displacements of x from the current peak position. The coefficients of this Taylor's expansion of the potential function V would give the quantities "slope" F and "curvature" $m\omega^2$ that enter into our differential equations for $\alpha(t)$ and $\beta(t)$. The α , β equations could then be integrated numerically for a short time step to give a new configuration expressed in terms of new values of $\alpha(t)$ and $\beta(t)$. From these, a new peak position x_p for the later time could be calculated. After this, a new Taylor's expansion could be made about the new peak location x_p , where new values for F and ω^2 would be calculated, and so on. The process is similar to that used in analytic continuation in the theory of functions of a complex variable.

I list a few more problems which may be treated in later publications.

- (4) a freely swinging simple pendulum (or anharmonic oscillator);
- (5) a periodically forced anharmonic oscillator (Duffing³⁰ problem),
- (6) a Kepler problem, as in the motion of a comet headed directly at the sun, where the potential energy varies with the inverse first power of the distance between the comet and the sun;
- (7) a spinless charged particle moving in a magnetic field; and
- (8) a measurement problem in quantum mechanics and in quantum optics.

B. More general potential functions

So far, we have considered problems in one Cartesian coordinate x for a single particle. The method can easily be extended to apply to more complicated potentials and to two or three Cartesian coordinates for a single particle, or to any number of such particles. We begin by assuming that $V(x,y,z,\dots)$ has linear and quadratic terms in $x,y,z,\dots$, and also allow terms in $xy,xz,yz,\dots$. We merely have to devise a convenient notation for parameters to replace the $\alpha(t)$ and $\beta(t)$ of the earlier treatment. For one coordinate, one needs only two complex functions, $\alpha(t)$ and $\beta(t)$. For two degrees of freedom there are five. In the case of n degrees of freedom, there are

$$n + \frac{n(n+1)}{2} = \frac{n(n+3)}{2},$$

functions of time. The method analogous to analytic continuation is readily available as before. Hence, it is simple to treat the orbits of Kepler for the $1/r$ potential. The ψ function $\psi(x,t)$ of x at time t now becomes a function $\psi(x,y,t)$ in a two-dimensional coordinate space of x and y . At the cost of more numerical work, systems with many particles could be treated. Nowadays, a computer can make large systems easy to treat.

Other applications of this work are to find the ways in which Gaussian ψ functions spread and are deformed in anharmonic potentials, and to discuss chaotic phenomena and the quantum theory of measurement. For lack of time and space, I will have to pass up such topics in the present discussion.

Many problems in electromagnetic theory and quantum optics can be reduced to the dynamical problem of a system consisting of a large number of mechanical simple harmonic oscillators. Each oscillator coordinate is associated with the electric field of one of the normal modes of a cavity resonator. Hence, the method of complex Gaussian ψ functions can be applied to many problems in the quantum theory of radiation. In case you don't know it already, despite what many other people⁴⁰ think, THERE ARE NO PHOTONS OF A PARTICLE-LIKE NATURE IN QUANTUM ELECTRODYNAMICS.

XIV. EHRENFEST'S THEOREM REVISITED

Ehrenfest's theorem of 1927 showed that a sufficiently narrow ψ packet would follow the Newtonian classical trajectory for a problem with a potential energy $V(x)$. In a later paper, we will show that a Gaussian packet remains Gaussian as time advances if and only if the potential energy $V(x)$ is parabolic in its x dependence. If the potential is nonparabolic, a Gaussian ψ function will become non-Gaussian as time advances. The time required for this transition becomes longer in the classical limit, i.e., as $\hbar$ approaches zero. It seems that this behavior has not been previously noticed. Bruria Kaufman and I have developed a perturbation theory which permits convenient calculations of small departures from the classical limit, which may be published in a future paper.

XV. STATIONARY STATES

The ψ equation has many solutions that are not in the form of narrow packets. There are even solutions which are periodic in time,

$$\psi(x,t) = u(x) \exp\left(-\frac{iEt}{\hbar}\right),$$

with a circular frequency, $E/\hbar$, where E is a constant with dimensions of an energy. These are called stationary solutions, since their probability distribution functions $W(x,t) = |\psi(x,t)|^2$ do not depend on time. They would not have seemed to have any physical application before the time of Planck. They are the stationary states of the twentieth century. The concept of a space-time orbital trajectory naturally has no validity for such a state.

XVI. EXTENSIONS TO OTHER PROBLEMS: MAGNETIC FIELDS?

The above methods can also be applied to the motion of a charged particle in a spatially variable magnetic field [described by a vector potential (**A**)]. This case may also be treated in a future paper.

XVII. SPECIAL RELATIVITY

Instead of starting from Newton's classical equation of motion, we could try to start from Einstein's 1905 special relativistic equation of motion. This can be taken as $F = dp(t)/dt$ where the momentum $p(t) = [m(|v|)v]$, and F is the force acting on the particle of mass $m(|v|)$. The mass $m(|v|)$ depends on the (time-dependent) speed $|v|$ of the particle velocity v according to a well-tested law

$$m(|v|) = \frac{m}{\text{Sqrt}\left(1 - \left(\frac{v}{c}\right)^2\right)},$$

where m is the rest mass $m(0)$ of the particle.

In the Newtonian case, we dealt with the chaotic behavior of classical nonrelativistic mechanics and converted Newton's equation of motion $F = ma$ into super classical quantum mechanics and its nonrelativistic Schrödinger equation. The Einstein special relativistic equation of motion leads to even stronger chaotic features than does the Newtonian classical one. However, it does not seem to be possible to derive a satisfactory generalization of the Schrödinger equation for the Einstein 1905 special relativity theory. Historically, some candidates for such an equation have been considered, and rejected. One of these has second derivatives of ψ with respect to time. Another one attempts to work with space derivatives $\partial/\partial x$ imbedded in a square-root expression containing the momentum $p = [m(|v|)v]$. In the face of these difficulties, we can try to revert to the good old days of 1927, and follow Dirac in his introduction of a four-component wave equation. I never hoped that I was going to solve the problems of relativistic quantum mechanics in this paper.

XVIII. SPIN?

Consider the motion of a particle in special relativity mechanics. In the cases treated so far, the quantum theory has a classical, i.e., a nonquantum, limit. However, the Dirac relativistic theory does not have a simple WKB approximation,⁴¹ or classical limit. In fact, the case of a particle with a spin magnetic moment cannot be treated by the simple methods of this paper. I have been considering the theory of a spinless particle. Spin phenomena are very important for chemistry and physics. For the conventional form of quantum mechanics, a Pauli two-component probability amplitude function is easily incorporated into the nonrelativistic quantum mechanics. The magnetic moment associated with spin is of some considerable interest, but usually does not dominate the problems of chemical binding. However, the spin itself plays a vital role in the theory of chemical binding, in the proper enumeration of states, and the whole structure of the periodic system of elements. A failure of nature to get that right might literally be a matter of life and death for us. It is much more

difficult to extend the theory of probability distribution functions to particles with spin, either in the Pauli approximation or with the Dirac relativistic electron theory.

Historically, a nuclear atom with orbiting and spinning electrons seemed very similar to a planetary system with spinning planets in orbital motion about a star. However, in classical theory, the spinning planet is a many-body system. The Dirac relativistic wave equation was intended to be a one-particle theory, but it was unexpectedly forced to become a many-body (or field) theory because of the existence of negative energy states in relativistic quantum mechanics. It is possible, but fairly complicated, to apply the methods of this paper to a rotating object consisting of several particles. However, the spin of the electron is not treated in quantum mechanics in anything like that manner, but rather with multicomponent wave functions.

XIX. EXCHANGE DEGENERACY

Another interesting problem concerns particle exchange phenomena. We easily extended our treatment from a one-body to a many-body problem. A suitable generalization of quantum mechanics is obtained, but nothing is being said about the particle exchange symmetry of the ψ function. It is easily possible to impose Fermi–Dirac or Bose–Einstein statistics as it was done in the 1920s. We are thereby no worse off, then, than we were before. The Newtonian classical theory from which we start contains no hints for dealing with the problem of particle exchange. Berry⁴² has recently discussed similar problems from a different point of view.

XX. REMARKS ON SPIN MAGNETIC MOMENTS

The magnetic moment of the electron could be associated with a very rapid small-scale circular motion of the electron. The magnetic moment μ associated with a point charge e moving in a circle of radius R with a speed v is $\mu = eR(v/2)$. If R is taken to be the Dirac length $\hbar/mc$, note that R is about $\frac{1}{137}$ of the Bohr radius a_0 . One finds $\mu = (e\hbar/mc)(v/c)$, so that to obtain a magnetic moment of one Bohr magneton, $\mu_0 = e\hbar/mc$, one would have to have $v = c$.

This argument is not meant to be quantitative, but just shows that a spinning electron is a strange thing. If the electron were not point-like, but string-like, with a distributed line charge with a length of order $\hbar/mc$, the theoretical energy levels of the hydrogen atom would be very different from the observed values.

The trouble is that the simple theory is built on the concept of point particles which can, in principle, be considered to be isolated. However, an electron is inevitably in interaction with the quantum electromagnetic field. As suggested by Welton,⁴³ this means that the position of the overidealized point electron is effectively smeared out over a distance an order of magnitude larger than the classical electron radius. The resulting departure from the Coulomb inverse square law force contributes, in large part, to the electromagnetic level shift.

Nuclei also have a finite size. A proton has turned out to be a highly complex particle. At present, it seems to be that the most reliable measure of the proton size comes from studies of the atomic hydrogen levels.⁴⁴

XXI. IN RETROSPECT

The most obvious success of our theory is in the field of nonrelativistic quantum mechanics, and the very clear understanding of the statistical interpretation of quantum theory it gives in that domain. The so-called paradoxes of traditional quantum mechanics are *all* to be found in the nonrelativistic domain. The most obvious failure involves spin. The easiest way to include spin would be to imitate the Pauli–Dirac methods of multicomponent wave function. However, Gaussian exponential factors play a big role in our work, and the four components of a Dirac relativistic wave function have different exponential factors. The Gaussian exponential factors do not cancel out as nicely as they do in the nonrelativistic theory.

It would have been much better for a rational history of physics if Newton had discovered probability theory, and applied it to the laws of his classical mechanics. A similar remark can be made about Einstein: He was probably the first physicist to make use of probability theory to discuss a physical problem in Newtonian mechanics, i.e., Brownian motion. Unfortunately, Einstein did not believe that the use of probability theory was essential for radioactive decay,^{45,46} and other microscopic problems, and not merely a convenient recipe for the avoidance of tedious details.

One can ask what Einstein would have thought about my probabilistic Newtonian mechanics. It is certainly known that he had great respect for NCM. In the limit of a very narrow probability distribution, the probabilistic function $W(x,t)$ would give exactly the same features as the trajectory $x = x(t)$ of NCM as shown in my discussion of wave packet propagation in the classical limit.

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¹W. E. Lamb, Jr., “On the electromagnetic properties of nuclear systems,” *Phys. Rev.* **53**, 651–656 (1938).

²W. E. Lamb, Jr., “Suppose Newton had invented wave mechanics,” *Am. J. Phys.* **62**, 201–206 (1994).

³W. E. Lamb, Jr., “The borderland between quantum and classical mechanics,” *Phys. Scr.* **T70**, 7–13 (1997).

⁴W. E. Lamb, Jr., “An operational interpretation of nonrelativistic quantum mechanics,” *Phys. Today* **22** (4), 23–28 (1969).

⁵J. von Neumann, *Mathematical Foundations of Quantum Mechanics*, English translated by R. T. Beyer (Princeton U.P., Princeton, NJ, 1955).

⁶W. E. Lamb, Jr., “Von Neumann’s reduction of the wave function,” in *The Centrality of Science and Absolute Values, Proceedings of the Fourth International Conference on the Unity of the Sciences* (ICF, New York, 1975), pp. 297–303.

⁷W. E. Lamb, Jr., “Remarks on the interpretation of quantum mechanics,” in *Science of Matter*, edited by S. Fujita (Gordon and Breach, New York, 1979), pp. 1–8.

⁸W. E. Lamb, Jr., “A letter to Kip Thorne,” in *CCNY Physics Symposium, in Celebration of Melvin Lax’s Sixtieth Birthday*, edited by H. Falk (City University of New York, New York, 1983), pp. 40–43.

⁹E. Arthurs and J. Kelly, “On the simultaneous measurement of a pair of conjugate observables,” *Bell Syst. Tech. J.* **44**, 725–729 (1965).

¹⁰W. E. Lamb, Jr., “Quantum theory of measurement,” *Ann. N.Y. Acad. Sci.* **480**, 407–416 (1986).

¹¹W. E. Lamb, Jr., “Theory of quantum mechanical measurements,” in *Proceedings of the 2nd International Symposium on the Foundations of Quantum Mechanics: In the Light of New Technology*, edited by M. Namiki (Japan Physical Society, 1987), pp. 185–192.

¹²W. E. Lamb, Jr., “Sequential measurements in quantum mechanics,” Lecture at Como NATO Advanced Research Workshop in Quantum Optics, September, 1986, in *Quantum Measurement and Chaos*, edited by E. R. Pike and S. Sarkar (Plenum, New York, 1987), pp. 183–193.

¹³W. E. Lamb, Jr., “Classical measurements on a quantum mechanical system,” in *Proceedings of the International Symposium on Spacetime Symmetries*, edited by Y. S. Kim and W. M. Zachary (North-Holland, Amsterdam, 1987), pp. 1–10.

- dam, 1989), pp. 197–201 [reprinted from Nucl. Phys. B (Proc. Suppl.) **6** (1989)].
- ¹⁴W. E. Lamb, Jr., “Quantum theory of measurement,” in *Lecture at Turin NATO Advanced Research Workshop on Noise and Chaos in Nonlinear Dynamical Systems*, March, 1989, edited by F. Moss, L. Lugiato, and W. Schleich (Cambridge U.P., Cambridge, UK, 1990), pp. 1–14.
- ¹⁵*Quantum Theory and Measurement*, edited by J. A. Wheeler and W. H. Zurek (Princeton U.P., Princeton, NJ, 1983).
- ¹⁶S. Goldstein, “Quantum theory without observers,” *Phys. Today* **51**, 42–47 (1998); **51**, 38–42 (1998); *ibid.* **52**, 11 (1999); A. Zeilinger, “Experiment and the foundation of quantum physics,” pp. 482–498 in *More Things in Heaven and Earth, A Celebration of Physics at the Millennium*, edited by B. Bederson (Springer, New York, 1999); a conference, overloaded with such ideas, was recently held at Lake Garda, Italy, September (2000) (unpublished).
- ¹⁷See also, J. G. Cramer, “Transactional interpretation of quantum mechanics,” *Rev. Mod. Phys.* **58**, 647–688 (1986); See also, G. Auletta, *Foundations and Interpretation of Quantum Mechanics* (World Scientific, Singapore, 2000).
- ¹⁸A. Einstein, “Zur quantentheorie der strahlung,” *Phys. Z.* **18**, 121–128 (1917); See, however, W. E. Lamb, Jr., “Laser theory and Doppler effect,” *IEEE J. Quantum Electron.* **QE-20**, 551–554 (1984).
- ¹⁹See the Schrödinger cat paradox, in *Quantum Theory and Measurement*, edited by J. A. Wheeler and W. H. Zurek (Princeton U.P., Princeton, NJ, 1983), pp. 152–167.
- ²⁰W. E. Lamb, Jr., “Schrödinger’s Cat,” in *Paul Adrian Maurice Dirac*, edited by B. N. Kursunoglu and E. P. Wigner (Cambridge U.P., Cambridge, UK, 1987).
- ²¹G. Auletta, “Delayed choice experiments,” *Foundations and Interpretation of Quantum Mechanics* (World Scientific, Singapore, 2000), Chap. 26, footnote 17.
- ²²W. E. Lamb, Jr., “Comments by W. E. Lamb,” in *Workshop on Quantum Cryptography and Quantum Computing*, Proceedings, February 15, 1995, edited by H. Everitt (U.S. Army Research Office, 1995), pp. 24 and 25.
- ²³M. Born, “On the Interpretation of Quantum Mechanics,” in *Les Prix Nobel* (Stockholm, 1954).
- ²⁴W. E. Lamb, Jr., and H. Fearn, “Classical Theory of measurement: A big step toward the quantum theory of measurement,” in *Amazing Light*, a volume dedicated to Charles Hard Townes on his 80th Birthday, edited by R. V. Chiao (Springer-Verlag, New York, 1996).
- ²⁵See a biography of Maria Gaetana Agnesi, and also an article on special mathematical functions in recent editions of the *Encyclopedia Britannica*. A Lorentzian curve is often confused with the Witch of Agnesi.
- ²⁶J. G. Crowther, “Maxwell’s inaugural lecture,” *New Scientist Sci. J.* 478–481 (1971).
- ²⁷H. Poincaré, in *New Methods of Celestial Mechanics*, edited by D. L. Goroff (AIP, Woodbury, NY, 1993), Vols. 1–3.
- ²⁸W. E. Lamb, Jr., “Quantum chaos and the theory of measurement,” in *Chaotic Behavior in Quantum Systems. Theory and Applications*, edited by G. Casati, NATO Advanced Research Workshop on Quantum Chaos, Como, June, 1983 (Plenum, New York, 1985), pp. 253–261.
- ²⁹J. Lighthill, “The recently recognized failure of predictability in Newtonian dynamics,” *Proc. R. Soc. London, Ser. A* **407**, 35–50 (1936).
- ³⁰G. Duffing, *Forced Vibrations at Various Frequencies and their Technical Significance* (Vieweg, Braunschweig, 1918), pp. 41 and 42.
- ³¹E. Fermi, J. R. Pasta, and S. Ulam, “Studies of nonlinear problems,” in Los Alamos Report No. LA-1940 (1955).
- ³²M. Toda, *J. Phys. Soc. Jpn.* **21**, 431 (1967).
- ³³M. Born, Waynflete Lecture, Magdalen College, Oxford (1949) (unpublished).
- ³⁴P. Ehrenfest, “Notes on the approximate validity of quantum mechanics,” *Z. Phys.* **45**, 455–457 (1927).
- ³⁵L. I. Schiff, *Quantum Mechanics*, 3rd ed. (McGraw-Hill, New York, 1955), pp. 178–182.
- ³⁶E. Schrödinger, *Collected Papers on Wave Mechanics* (Blackie, London, 1928).
- ³⁷E. H. Kennard, “The quantum mechanics of simple types of motion,” *Z. Phys.* **44**, 326–352 (1927).
- ³⁸R. J. Glauber, “Coherent and incoherent states of the radiation field,” *Phys. Rev.* **131**, 2766–2788 (1963).
- ³⁹M. Sargent, M. O. Scully, and W. E. Lamb, Jr., “Laser Physics,” in Appendix H, *The Coherent State* (Wiley, New York, 1977), Appendix H. This is largely based on Kennard’s approach. Many results are simpler to get using the complex Gaussian method of this paper.
- ⁴⁰W. E. Lamb, Jr., “Antiphoton,” *Appl. Phys. B: Lasers Opt.* **60**, 77–84 (1995).
- ⁴¹J. Bolte and S. Keppeler, “Semiclassical time evolution and trace formula for relativistic spin- $\frac{1}{2}$ particles,” *Phys. Rev. Lett.* **81**, 1987–1990 (1998).
- ⁴²M. V. Berry, From a Lecture in May, 1999, at the University of Arizona (unpublished).
- ⁴³T. A. Welton, “Some observable effects of the quantum mechanical fluctuation of the electromagnetic field,” *Phys. Rev.* **74**, 1157–1167 (1948).
- ⁴⁴S. Bourzeix *et al.*, “High resolution spectroscopy of the hydrogen atom: Determination of the 1S Lamb shift,” *Phys. Rev. Lett.* **76**, 384–387 (1996).
- ⁴⁵See Einstein’s Remarks on Radioactive Decay in *Albert Einstein: Philosopher-Scientist*, edited by P. A. Schilpp (Open Court, La Salle, IL, 1970), pp. 665–688.
- ⁴⁶See E. T. Whittaker, *History of the Theories of the Aether and Electricity* (Nelson, London, 1951–1953).

SCIENTISTS’ RESPONSIBILITY

The grave responsibility of scientists is to use their voices to blow back the fog that shrouds the minds of those who have not yet seen. Scientists are successfully treading the path of reductionism. They are exposing the simple essentials of the world, seeing its mechanism, seeing that they can comprehend its actions, and seeing that they can understand its origin and elucidate the problems that have puzzled people and given priests their power.

P. W. Atkins, “The Limitless Power of Science,” in *Nature’s Imagination—The Frontiers of Scientific Vision*, edited by John Cornwell (Oxford University Press, New York, 1995).